

Townsend 1.4-1.6

1.4)

For general solutions of complex amplitudes, we have a phase $e^{-i\delta_{\pm}}$ $|c_{\pm}|$.

Expectation value is

$$\langle S_z \rangle = c_+^* c_+ \left(\frac{\hbar}{2} \right) + c_-^* c_- \left(-\frac{\hbar}{2} \right)$$

$$(\Delta S_z)^2 = \langle s_z^2 \rangle - \langle s_z \rangle^2$$

1.5

Relative phases are all that matter, use a right handed coordinate system.